

# Recombination Rate Coefficients in EXHALE: H I, He I, He II, and the He I $2^3S$ Triplet, with Benchmarks and the Status of Newer Data

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## 1. Overview

This memo documents how EXHALE treats recombination onto H I, He I, He II, and the metastable He I  $2^3S$  level (the lower level of the He 10830 line), benchmarks the adopted coefficients against other published fits, and reviews whether more recent data warrant an update. The companion to this is the photoionization cross-section memo (photoion\_cross\_sections.tex); together they cover the two microphysical inputs that set the H/He ionization balance and the He  $2^3S$  population.

The two headline conclusions are: (i) EXHALE's H/He *case-B* coefficients (Hui & Gnedin 1997) agree with the Draine (2011) case-B fits to  $\sim 2\%$  over  $5 \times 10^3 - 2 \times 10^4$  K — they are not outdated; the larger differences seen against some escape codes are the *case-A vs. case-B* modeling choice, not data vintage. (ii) EXHALE's He  $2^3S$  triplet recombination (Benjamin, Skillman & Smits 1999, as parametrized by Oklopčić & Hirata 2018) is *identical* to that adopted by the most recent multi-species escape model (Taylor et al. 2025), so it reflects current escape-community practice.

## 2. What EXHALE uses

All H/He recombination coefficients are defined in `src/modules/radiation/Cool_coeff.f90` and assembled in `ionization_equilibrium.f90`. Units are  $\text{cm}^3 \text{s}^{-1}$ .

**H II → H I and He III → He II (Hui & Gnedin 1997, case B).** With  $\lambda_i = 2T_i/T$  and  $T_i$  the ionization temperature (157807, 285335, 631515 K for H I, He I, He II),

$$\alpha_{\text{H II}}^B(T) = 2.753 \times 10^{-14} \frac{\lambda_{\text{H}}^{1.5}}{[1 + (\lambda_{\text{H}}/2.74)^{0.407}]^{2.242}}, \quad (1)$$

$$\alpha_{\text{He III}}^B(T) = 2 \times 2.753 \times 10^{-14} \frac{\lambda_{\text{He}^+}^{1.5}}{[1 + (\lambda_{\text{He}^+}/2.74)^{0.407}]^{2.242}}, \quad (2)$$

the second being the hydrogenic ( $Z = 2$ ) scaling of the first. Case B (the “on-the-spot” approximation) excludes direct recombination to the ground state, whose re-emitted photon is assumed re-absorbed locally.

**He II → He I (two regimes).** When the He 2<sup>3</sup>S network is *off*, EXHALE uses the Hui & Gnedin (1997) total case-B coefficient  $\alpha_{\text{He II}}^B = 1.26 \times 10^{-14} \lambda_{\text{He}}^{0.75}$ . When the network is *on*, the He II → He I recombination is split into singlet and triplet channels (ionization\_equilibrium.f90; the total case-B value is overwritten), following Benjamin, Skillman & Smits (1999) as parametrized by Oklopčić & Hirata (2018):

$$\alpha_1(T) = 1.54 \times 10^{-13} (T/10^4)^{-0.486} \quad (\text{singlet, } 1^1S), \quad (3)$$

$$\alpha_3(T) = 2.10 \times 10^{-13} (T/10^4)^{-0.778} \quad (\text{triplet, } 2^3S). \quad (4)$$

These are *effective* (cascade-summed) coefficients:  $\alpha_3$  is the total rate at which recombinations populate the metastable 2<sup>3</sup>S level via direct capture plus radiative cascades through the higher triplet terms. The triplet captures the majority of He recombinations,  $\alpha_3/(\alpha_1 + \alpha_3) \simeq 0.53\text{--}0.63$  over  $5 \times 10^3\text{--}2 \times 10^4$  K (Table 2).

Metals use the modern Badnell radiative + dielectronic recombination (RR+DR), described in the cooling memo, and are not revisited here.

### 3. H<sup>+</sup> → H: case-A vs. case-B, not data vintage

Figure 1 and Table 1 compare EXHALE’s case-B  $\alpha_{\text{H II}}$  with the Draine (2011) case-A/B fits and with the total (case-A) coefficient adopted by the recent escape model of Taylor et al. (2025, their Storey & Hummer 1995 value). EXHALE tracks the Draine *case-B* curve to  $\sim 2\%$  across the whole range. The Taylor+2025 value is  $\sim 40\text{--}90\%$  higher, but that is because it is a *case-A* (total) rate: codes that transport the diffuse recombination continuum explicitly use case A, whereas EXHALE’s on-the-spot scheme correctly uses case B. The underlying data agree; the difference is the radiative-transfer treatment.

Table 1: H<sup>+</sup> → H recombination coefficient ( $10^{-13} \text{ cm}^3 \text{ s}^{-1}$ ).

| $T$ [K] | EXHALE (B) | Draine (B) | Draine (A) | Taylor+25 (A) |
|---------|------------|------------|------------|---------------|
| 5000    | 4.543      | 4.428      | 6.733      | 6.608         |
| 10000   | 2.592      | 2.540      | 4.130      | 4.241         |
| 15000   | 1.837      | 1.818      | 3.087      | 3.271         |
| 20000   | 1.428      | 1.428      | 2.505      | 2.721         |

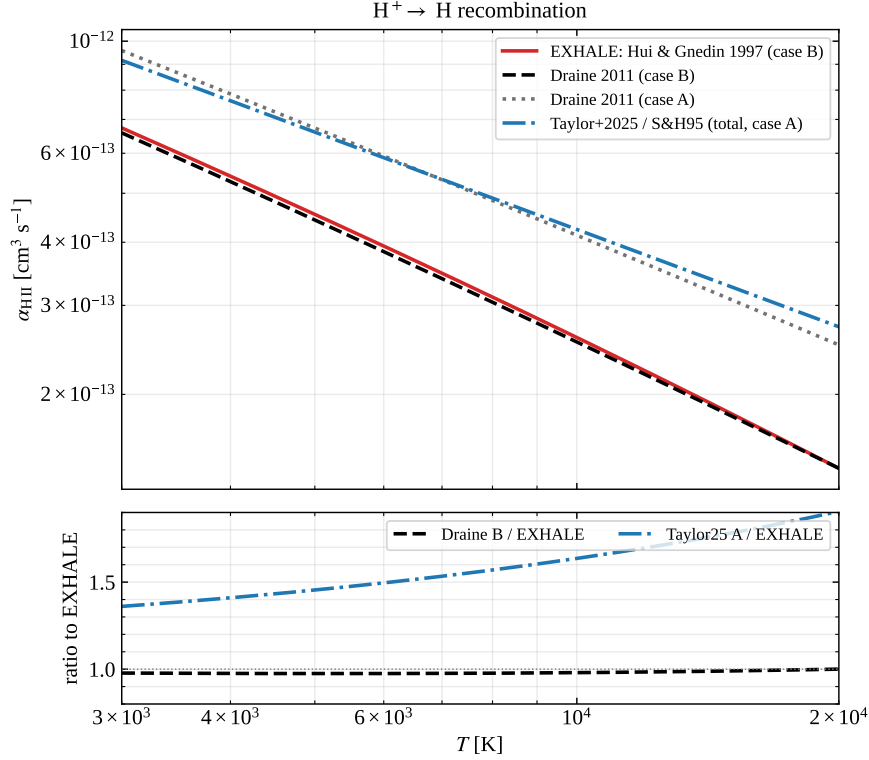


Figure 1:  $H^+ \rightarrow H$  recombination. EXHALE (Hui & Gnedin 1997, case B; red) overlaps the Draine (2011) case-B fit to  $\sim 2\%$  (bottom panel); the case-A (total) curves of Draine and Taylor+2025 are  $\sim 40\text{--}90\%$  higher. The offset is the case-A vs. case-B choice, not a data update.

## 4. Helium recombination

The left panel of Fig. 2 repeats the case-A/B story for  $He^{++} \rightarrow He^+$ : EXHALE (case B) matches the hydrogenic Draine case-B to  $\sim 2\%$ , while the Verner & Ferland (1996) total used by Taylor+2025 is  $\sim 45\%$  higher (case A). The right panel shows  $He^+ \rightarrow He$ : the Hui & Gnedin total case-B, and the Benjamin+1999 singlet/triplet split that EXHALE uses when the 2<sup>3</sup>S network is active. Table 2 lists the values.

Two points are worth noting. (i) The triplet recombination  $\alpha_3$  (the quantity that feeds He 10830) is *identical* in EXHALE and in the latest escape models — both adopt Benjamin+1999. (ii) The Benjamin+1999 effective total  $\alpha_1 + \alpha_3$  is  $\sim 40\%$  larger than the Hui & Gnedin (1997) total case-B (e.g.  $3.64$  vs.  $2.62 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$  at  $10^4 \text{ K}$ ); the two come from different methods (effective level-resolved cascade sums vs. a recombination-spectrum case-B total). When the 2<sup>3</sup>S network is on, EXHALE consistently uses the Benjamin split for both channels, so the He ionization balance and the metastable source term are computed from the same data set.

## 5. Is there newer data?

**H I and He II (hydrogenic).** Essentially solved. Hui & Gnedin (1997), Storey & Hummer (1995), Verner & Ferland (1996), and Badnell (2006) agree to  $\sim 1\text{--}2\%$  (Fig. 1). No update is warranted; the only meaningful choice is case A vs.

Table 2: He recombination coefficients ( $10^{-13} \text{ cm}^3 \text{ s}^{-1}$ ). Left:  $\text{He}^{++} \rightarrow \text{He}^+$ . Right:  $\text{He}^+ \rightarrow \text{He}$  — Hui & Gnedin total (case B) and the Benjamin+1999 singlet  $\alpha_1$ , triplet  $\alpha_3$ , and triplet fraction.

| $T$ [K] | $\text{He}^{++} \rightarrow \text{He}^+$ |           |           | $\text{He}^+ \rightarrow \text{He}$ |            |            |                                |
|---------|------------------------------------------|-----------|-----------|-------------------------------------|------------|------------|--------------------------------|
|         | EXHALE(B)                                | Draine(B) | Tayl25(A) | HG97 tot                            | $\alpha_1$ | $\alpha_3$ | $\alpha_3/(\alpha_1+\alpha_3)$ |
| 5000    | 25.61                                    | 25.35     | 34.51     | 4.400                               | 2.157      | 3.601      | 0.625                          |
| 10000   | 15.45                                    | 15.13     | 22.61     | 2.616                               | 1.540      | 2.100      | 0.577                          |
| 15000   | 11.37                                    | 11.09     | 17.66     | 1.930                               | 1.265      | 1.532      | 0.548                          |
| 20000   | 9.09                                     | 8.86      | 14.82     | 1.556                               | 1.100      | 1.225      | 0.527                          |

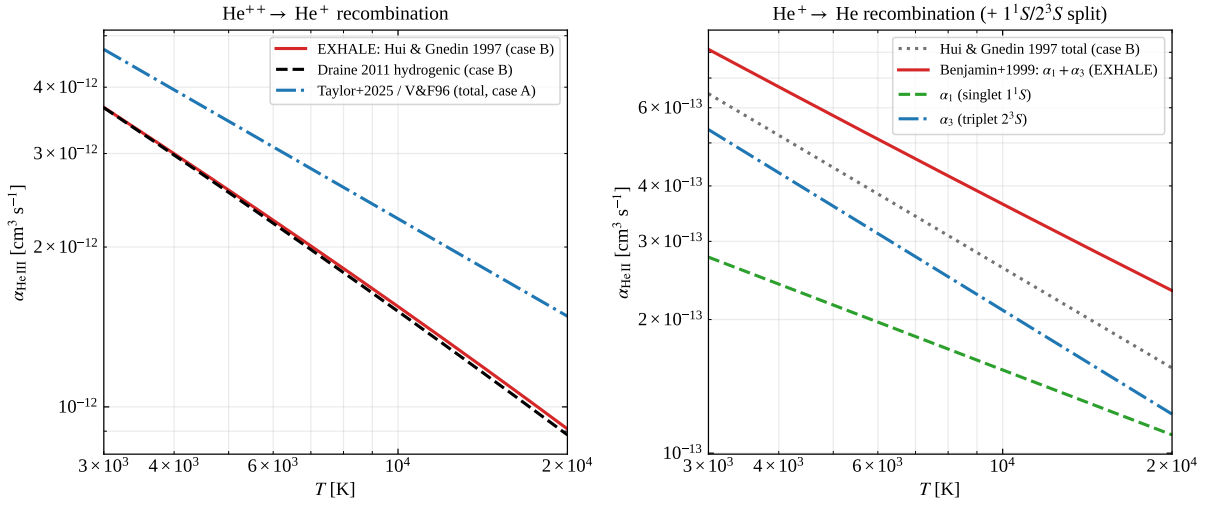


Figure 2: Helium recombination. **Left:**  $\text{He}^{++} \rightarrow \text{He}^+$  — EXHALE case B (red)  $\approx$  Draine hydrogenic case B (black); Verner & Ferland 1996 / Taylor+2025 total (case A) is  $\sim 45\%$  higher. **Right:**  $\text{He}^+ \rightarrow \text{He}$  — Hui & Gnedin total case B (dotted) and the Benjamin+1999 singlet/triplet split ( $\alpha_1 + \alpha_3$ ,  $\alpha_1$ ,  $\alpha_3$ ) used by EXHALE when the 2<sup>3</sup>S network is active.

B (§3.).

**He I total and the 2<sup>3</sup>S triplet.** The data EXHALE uses (Benjamin, Skillman & Smits 1999) have been superseded *in the nebular community* by the improved He I effective recombination coefficients of **Porter, Ferland, Storey & Detisch (2012, MNRAS 425, L28; 2013 erratum MNRAS 433, L89)** — now the default in Cloudy — and more recently by **Del Zanna & Storey (2022)**. These differ from Benjamin+1999 at roughly the 5–15% level over the relevant temperatures.

However, the exoplanet *escape* community has *not* migrated: the most recent multi-species escape model (Taylor et al. 2025) still adopts Benjamin+1999 for the triplet — the same  $\alpha_3 = 2.10 \times 10^{-13} (T/10^4)^{-0.778} \text{ cm}^3 \text{ s}^{-1}$  that EXHALE uses — and notes that recombination coefficients adopted across escape studies still “vary by a factor of 2–3,” a spread dominated by the case-A/B choice and by whether the singlet/triplet split is used at all, not by the  $\sim 10\%$  Porter update.

EXHALE is therefore fully consistent with current escape-modeling practice.

**Recommendation.** For the present goals (reproducing He 10830 at the  $\sim 10\%$  level) the Benjamin+1999 triplet rate is adequate and matches the field. If sub-10% precision on the metastable population is ever required, the worthwhile refinement is to adopt the Porter+2012/2013 (or Del Zanna & Storey 2022) effective recombination coefficients for  $\alpha_1$  and  $\alpha_3$ , ideally with their density dependence; this would require tabulating their coefficients rather than the simple power-law fits used now. The H/He case-B rates should be left as they are.

## 6. Reproducing this comparison

```
cd ATES/EXHALE/docs
python3 compare_recombination.py
# -> figures/rec_H.pdf, figures/rec_He.pdf
```

The script ports the EXHALE coefficients from `Cool_coeff.f90` verbatim and codes the Draine (2011), Verner & Ferland (1996) / Taylor+2025, and Benjamin+1999 reference fits.